

# Project Title: Nimbus App

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### 1. Project Overview & Objective

This project aims to transform existing Python-based financial extraction scripts into an intuitive and interactive application hosted on AWS. The application ingests live financial data via Yahoo Finance to deliver corporate metrics, financial ratios, and qualitative insights.

### 2. Dataset

The application will pull live data from Yahoo Finance upon user input. Afterwhich, the application stores the data into a backend which may be used for analytics or future reference.

Data pulled from Yahoo Finance includes but not limited to:

1. Company stock price
2. Filings made by the company
3. Relevant news regarding the company

### 3. Proposed Methodology

We expect to use Streamlit for the front end and AWS to host the app.

### 4. Proposed Features and Output

The primary design principle is user orientation. The application assumes users possess keen business acumen (e.g., business managers, analysts, investors) but limited technical expertise.

Our proposed features are split between necessary and add-ons, depending on their impact on the business and usability.

#### Necessary features:

| Feature            | Description                                                                                                                                                |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UI                 | User friendly interface that non-technical users would be able to quickly understand without additional instruction                                        |
| Company comparison | Ability to compare companies side by side (assume 2 companies first, if time allows, we implement multiple comparison)                                     |
| Storage            | User wants to store past reports, hence each use of the application should be saved (or unless user presses a button) as a report for access in the future |
| Scalability        | Must be able to support multiple concurrent users                                                                                                          |

|                                      |                                                                                                        |
|--------------------------------------|--------------------------------------------------------------------------------------------------------|
| Data Freshness Indicator (Timestamp) | Show when financial information was last updated/pulled                                                |
| Source Attribution                   | Identify the source of the financial data                                                              |
| User Authentication                  | Login/password system                                                                                  |
| Executive Summary                    | Generate concise AI-powered highlights and summary blocks based on the selected financial information. |

#### Add-on features:

| Feature                   | Description                                                                                                           |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Customizable Home Page    | Ability to personalize the dashboard/homepage layout                                                                  |
| Favorite/Pinned Companies | Ability to save companies so they can loaded automatically                                                            |
| Trending Searches         | Track frequently searched companies or stocks to provide useful insights for management.                              |
| Customizable Filters      | Allow users to filter and identify companies based on selected criteria such as profitability, growth, and liquidity. |

## 5. Potential Challenges

**Time constraint:** Given the limited time, it might not be possible to have all the provided features in this proposal. As such our aim would be to maximise the number of necessary features and only tackle the add-on features if time permits.

**Incomplete data from Yahoo Finance:** Some companies may have missing, unavailable, or inconsistent data.

**AI reliability:** AI features in Executive Summary may not give meaningful insights due to misinterpretation or hallucination.